

Layer 1: data schema

Portfolio project: a fictional but functional military logistics tool. This layer is the foundation, no ML yet: structure, access control, vehicles with their inspection/repair history, weapons, ammunition, and stock.

How to read the tables

PK	Primary key: the unique identifier of each row in the table.
FK	Foreign key: a link to a row in another table (e.g. vehicle_id points to a vehicle).
INT / REAL / TEXT / DATE	Integer, decimal number, text and date.
empty	The field can be left unset (NULL, in database terms).
(1-3)	A 1-to-3 scale; the higher number is more critical or severe.

1. The tables, column by column

Column	Type	What it is	Example
STRUCTURE - who and where			
unit	<i>Org chart: divisions, brigades, battalions</i>		
id	INT PK	Unit identifier	3
name	TEXT	Unit name	34th Battalion
type	TEXT	Echelon: division, brigade, battalion	battalion
parent_unit_id	INT FK	Unit it reports to (builds the hierarchy). Empty at the top	2
warehouse	<i>Where material is physically stored</i>		
id	INT PK	Warehouse identifier	2
name	TEXT	Warehouse name	North Central Depot
unit_id	INT FK	Owning unit. Empty if it is a central depot	empty
item	<i>Catalog of everything that can be held in stock</i>		
id	INT PK	Item identifier	42
code	TEXT	Unique item code	BRK-0042
name	TEXT	Description	Front brake pads
type	TEXT	spare_part, consumable or serial_equipment. Lets weapons/ammo reuse the same catalog	spare_part
category	TEXT	Item family	brakes

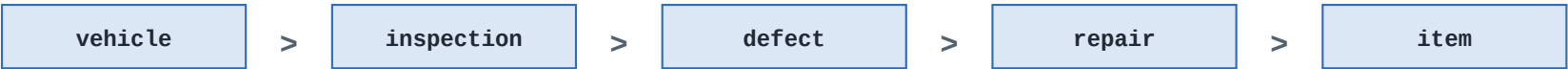
Column	Type	What it is	Example
criticality	INT (1-3)	How much its absence hurts readiness (3 = critical)	3
lead_time_days	INT	Days an order takes to arrive	45
ACCESS - who can see what			
app_user	<i>One row per person who logs in</i>		
id	INT PK	User identifier	5
username	TEXT	Login name	cmd.34bn
full_name	TEXT	Display name	Cpt. Alaoui
role	TEXT	admin (sees everything) or unit (scoped)	unit
unit_id	INT FK	Unit this user is scoped to. Empty for admin	3
VEHICLES - the history			
vehicle	<i>Each vehicle, tracked individually</i>		
id	INT PK	Internal identifier	317
serial_number	TEXT	Serial number / plate	VEH-00317
model	TEXT	Vehicle model	Light 4x4 truck
unit_id	INT FK	Unit it is assigned to	3
status	TEXT	active, in_repair or decommissioned	active
mileage_km	INT	Current mileage	61000
in_service_date	DATE	When it entered service	2019-03-12
inspection	<i>Each roadworthiness inspection (many per vehicle)</i>		
id	INT PK	Inspection identifier	908
vehicle_id	INT FK	Vehicle inspected	317
date	DATE	Day of the inspection	2025-01-15
mileage_km	INT	Mileage at that time (used by the model)	61000
result	TEXT	passed or defects_found	defects_found
defect	<i>Each defect found in an inspection</i>		
id	INT PK	Defect identifier	1500
inspection_id	INT FK	Inspection where it was found	908
system	TEXT	Brakes, lights, tires, engine, transmission, suspension, electrical, steering	brakes

Column	Type	What it is	Example
severity	INT (1-3)	Severity (3 = severe)	2
description	TEXT	Free text	Worn brake pads
repair	Each fix, linked to an inspection defect or standalone (breakdown between inspections)		
id	INT PK	Repair identifier	2210
vehicle_id	INT FK	Vehicle repaired	317
defect_id	INT FK	Defect being fixed. Empty for a standalone breakdown	1500
date	DATE	Day of the repair	2025-01-20
type	TEXT	part_replacement, adjustment or repair	part_replacement
item_id	INT FK	Part used (empty if none was used)	42
quantity	INT	Units of the part used	2
labor_hours	REAL	Workshop hours	1.5
outcome	TEXT	fixed or recurred (flags bad parts or failed repairs)	fixed
WEAPONS - serialized equipment			
weapon	Each weapon, tracked individually by serial number		
id	INT PK	Internal identifier	88
serial_number	TEXT	Serial number	RIF-04521
type	TEXT	Weapon type	assault_rifle
caliber	TEXT	Caliber	5.56mm
unit_id	INT FK	Unit it is assigned to	3
status	TEXT	active, in_repair or decommissioned	active
last_inspection_date	DATE	Last individual check	2025-06-01
rounds_fired	INT	Cumulative rounds fired (proxy for wear)	3200
AMMUNITION - lot-tracked consumable			
ammo_lot	Each production lot of ammunition, because it expires		
id	INT PK	Lot identifier	12
item_id	INT FK	Catalog item (e.g. 5.56mm ball)	77
lot_number	TEXT	Manufacturer's lot number	LOT-2023-118
manufacture_date	DATE	When it was produced	2023-05-10

Column	Type	What it is	Example
expiry_date	DATE	When it expires	2028-05-10
STOCK - what exists and what should exist			
allowance	<i>What each warehouse should hold</i>		
warehouse_id	INT FK	Warehouse	2
item_id	INT FK	Item	42
authorized_qty	INT	What that warehouse should hold	20
minimum_qty	INT	Below this, an alert fires	8
movement	<i>Everything that enters, leaves or moves. On-hand stock is calculated from this</i>		
id	INT PK	Movement identifier	70123
date	DATE	Day of the movement	2025-01-20
warehouse_id	INT FK	Warehouse where it happens (source in a transfer)	2
item_id	INT FK	Item moved	42
type	TEXT	receipt, issue, consumption, transfer or adjustment	consumption
quantity	INT	Units. Always positive; type decides +/- (adjustment can carry a sign)	2
dest_warehouse_id	INT FK	Transfers only: receiving warehouse	empty
repair_id	INT FK	Repair that caused the consumption, if any	2210
lot_id	INT FK	Ammo lot moved, if the item is lot-tracked. Empty otherwise	empty
order	<i>Material on order: in transit or already received</i>		
id	INT PK	Order identifier	55
warehouse_id	INT FK	Warehouse that placed it	2
item_id	INT FK	Item ordered	42
quantity	INT	Units ordered	30
order_date	DATE	When it was placed	2025-01-10
expected_date	DATE	Expected arrival (order date + lead time)	2025-02-24
received_date	DATE	When it actually arrived. Empty = still in transit	empty

2. How a vehicle's records connect

Each vehicle has many inspections, each inspection can have several defects, and each defect is fixed by one or more repairs that use an item from the catalog. Breakdowns between inspections are standalone repairs (no defect attached).



Every repair that uses a part creates a **movement** of type consumption, which is how the vehicle data and the stock data stay linked.

3. Example: one vehicle's history

Date	Record	Detail
Jan 2024	Inspection	42,000 km, defects found
Jan 2024	Defect	Brakes: worn pads
Jan 2024	Repair	Pads replaced (2 units), 1.5 h, fixed
Aug 2024	Standalone repair	Battery failure, replaced (1 unit), 0.5 h
Jan 2025	Inspection	61,000 km, passed

Each inspection, together with the vehicle's state at that moment, is a training example. The repairs that follow are the answer the model has to learn to predict: which part gets replaced, and when.

4. On-hand stock: no table for it

The stock of each item at each warehouse is calculated by summing **movements**. That way it never drifts out of sync with the movement log, and stock can be reconstructed for any past date, which is exactly what the model needs for training. It will be exposed as a database view.

5. Access control: who sees what

This falls out of the **unit** hierarchy already in the schema (parent_unit_id). Each app_user is scoped to one unit. The set of "visible units" for that user is the unit itself plus every unit below it in the tree, computed by walking parent_unit_id downward. Every table that carries a unit_id (vehicle, weapon) or belongs to a warehouse tied to a unit gets filtered to that set before being shown.

3rd Division (id 1, parent: none) 12th Brigade (id 2, parent: 1) 34th Battalion (id 3, parent: 2) <- app_user "cmd.34bn" is scoped here 35th Battalion (id 4, parent: 2)

Logged in as	Visible units	Sees
cmd.34bn (34th Battalion)	34th Battalion only	Only its own vehicles, weapons, warehouse and orders
cmd.12bde (12th Brigade)	12th Brigade + 34th Bn + 35th Bn	Its own data, plus both battalions' combined
cmd.3div (3rd Division)	Everything below it	The full division, rolled up
admin	All units	Everything, no filtering

This is authorization, not authentication: for a portfolio demo, a simple "log in as..." selector is enough, since there is no need for real passwords. The filtering logic is what matters and what a recruiter would actually look at.